8.3 Volume of Prisms and Pyramids

Name: ANSWERS

Part A: Introduction: In the design of containers and packages, two of the most important measurements to consider are:

volume and surface area.

The Volume of a three-dimensional object is a measure of how much space it occupies.

Volume is measured in cubic units, because volume is a three-dimensional measurement.
Typical cubic units for volume are:

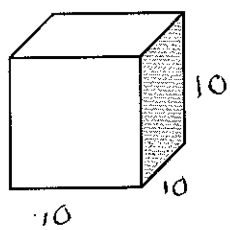
Cubic meters (m^3)

Cubic centimeters (cm^3)

Cubic millimeters (mm^3)

Another unit that is commonly used is the Liter (L), which is defined as the volume of a cube with sides of length 10cm.

1 Litre = 1000 **cm^3**



Litres are generally used when measuring the volume of a liquid. They are often used to describe the capacity of a container.

Capacity is the greatest volume that a container can hold. For example, your family may have a milk jug with a capacity of 1L.

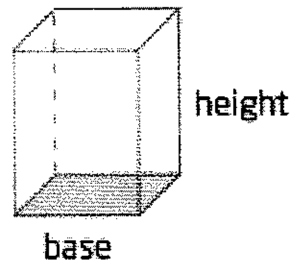
To go from cm^3 to Litres you must: ÷100

To go from Litres to cm^3 you must: ×1000

Part B: Volume of Prisms

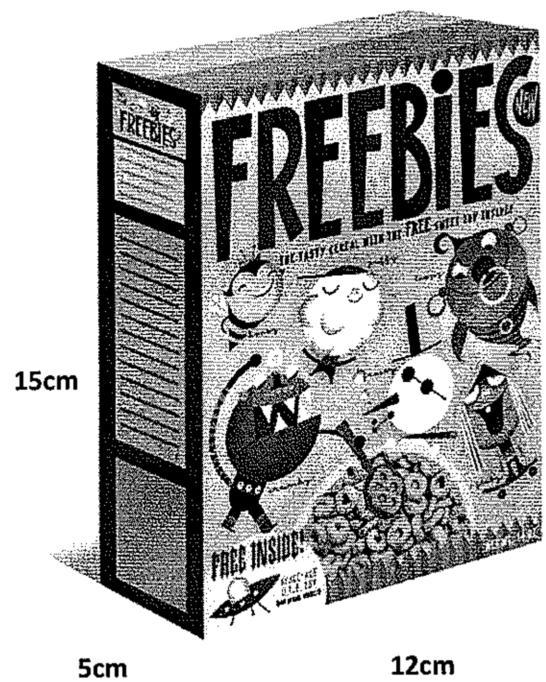
A prism is a 3-dimensional object with two parallel, congruent polygonal baes. A prism is named by the shape of its base. For example: rectangular prism, and triangular prism.

$$\textit{Volume of a prism} = (\textit{Area of base})(\textit{height})$$



Example 1: Volume of a Rectangular Prism

a) Determine the volume of the package in the photograph, in cubic centimetres.
b) Express the capacity in litres.



Solution:

a) $V = lwh$

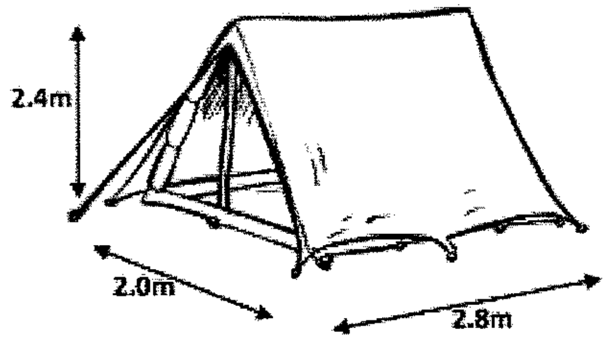
$= (5)(12)(15)$

$= 900\text{ cm}^3$

b) $900\text{ cm}^3 \times \dfrac{1\text{ L}}{1000\text{ cm}^3} = 0.9\text{ L}$

Example 2: Volume of a Triangular Prism

Determine the volume of the tent.



When finding the volume of a triangular prism you can Use the formula:

$$\boldsymbol{Volume = \frac{1}{2}blh}$$

b *stands for the bases of the triangular base*
l *stands for the height of the triangular base*
h *stands for the height of the prism*

Remember: $\boldsymbol{Volume\ of\ a\ prism = (area\ of\ base)(height)}$

Solution:

$V = \frac{1}{2}blh$

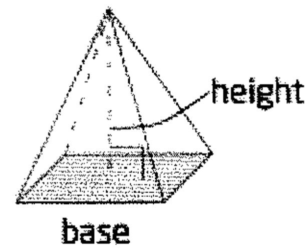
$= \frac{1}{2}(2)(2.4)(2.8)$

$= 6.72\,m^3$

Part C: Volume of Pyramids

A pyramid is a polyhedron whose base is a polygon and other faces are triangles that meet at a common vertex.

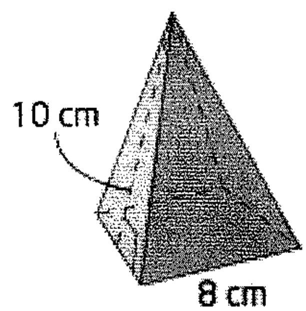
$$Volume\ of\ a\ pyramid = \frac{1}{3}(area\ of\ base)(height)$$



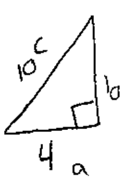
Example 3: Volume of a pyramid

Determine the volume of the pyramid-shaped container to the nearest cubic centimeter:

Remember: $Volume = \frac{1}{3}(area\ of\ base)(height)$



First determine the height of the pyramid using **Pythagorean theorem**: $a^2 + b^2 = c^2$

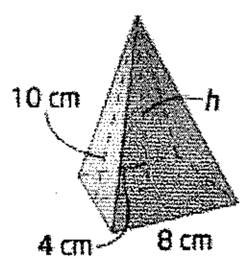


$b^2 = c^2 - a^2$

$b^2 = 10^2 - 4^2$

$b^2 = 84$

$b = 9.2cm$



The height of a pyramid is the perpendicular distance from the vertex to the base.

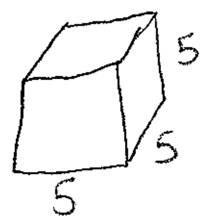
Now that you have the height of the pyramid, you can find the volume:

$$V = \frac{1}{3}(8^2)(9.2)$$
$$= 196.3\ cm^3$$

Part D: Show Me What You've Learned

1) A cube that has a side length of 5cm:

$$V = s^2$$
$$V = (5)^3$$
$$V = 125\ cm^3$$



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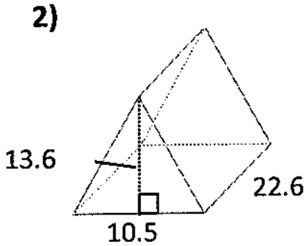
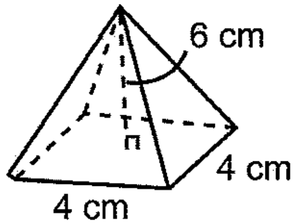


$$V = \frac{1}{2}blh$$
$$= \frac{1}{2}(10.5)(13.6)(22.6)$$
$$= 1613.6\ units^3$$

3)

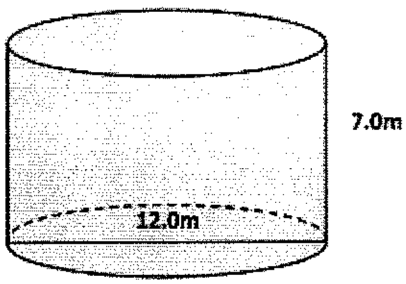


$V = \frac{1}{3}(\text{area of base})(\text{height})$

$= \frac{1}{3}(4^2)(6)$

$= 32 cm^3$

4)



Note: $Volume\ of\ a\ cylinder = \pi r^2 h$

$V = \pi r^2 h$

$= \pi(6^2)(7)$

$= 791.7\ m^3$

Part E: Conclusion

The volume of a prism is: (area of base)(height)

The volume of a pyramid is: $\frac{1}{3}$(area of base)(height)

Homework: Complete Worksheet